

Designing Empirical Research on Political Networks

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What is a network

- ▶ A network is a collection of things that are related to one another in one or more ways.

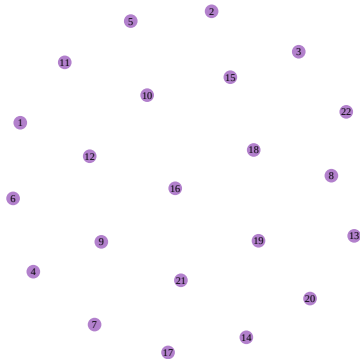
$$N = \{1, 2, \dots, n\}$$

E is the set of links (edges) between

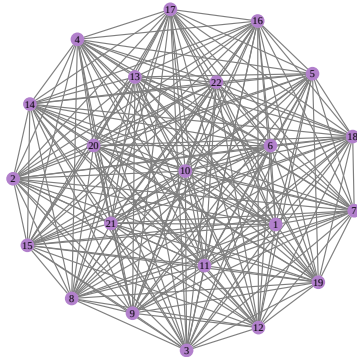
Link in network g between nodes i and j :

What is a social network

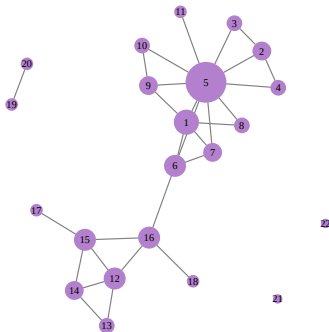
- ▶ In a social network, the nodes (things) are typically people, groups, organizations and the links (relationships) between them are social interactions like friendships, financial exchanges.



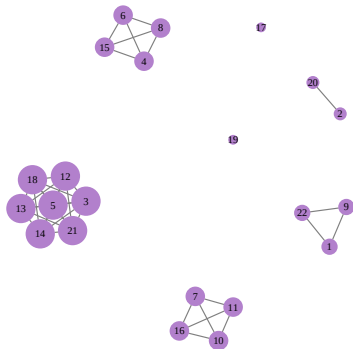
22 workshop participants



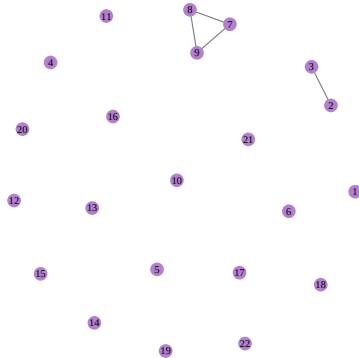
Nodes = workshop participants, link = workshop attendance



Nodes = workshop participants, size $\propto$ degree, link = interaction during coffee break



Nodes = workshop participants, size $\propto$ degree, link = interaction during lunch



Nodes = workshop participants, link = pre-existing relationship (classmates, colleagues)



The same set of nodes can be connected in different ways and form different social networks.

- ▶ a stranger
- ▶ your coffee shop barista
- ▶ your best friend
- ▶ someone you attended this workshop with
- ▶ someone you had coffee with at this workshop
- ▶ someone you had lunch with at this workshop
- ▶ someone you have a pre-existing tie with (friend, colleague) and who attended the workshop
- ▶ someone who had coffee with a person you had coffee with at this workshop
- ▶ a friend of a friend
- ▶ your mentor
- ▶ your mayor

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- ▶ your mentor
- ▶ your mayor

- ▶ Not all links matter for a given outcome.
- ▶ How two people are related may matter.
- ▶ How everyone else is related may matter.

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What is a network theory

- ▶ A well-specified network theory defines three elements (Larson 2024):
 - ▶ the set of nodes;
 - ▶ the types of links;
 - ▶ the function of the links.

Element	Question to Ask	Examples
Nodes	What is the set of things (actors, entities) that are connected?	Voters; politicians; organizations; committees; rebel groups; countries
Link Type	What is the relevant connection between two nodes?	Friendship; trust; dependence; kinship; shared membership; past interaction; collaboration
Link Function	What does the link do? Why is its presence theoretically important?	Transmits information; conveys influence or pressure; shares resources; aligns interests

Adapted from Larson (2024), Table 4.2.

- ▶ Nodes: ?
- ▶ Link types: ?
- ▶ Link functions: ?

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Political polarization is higher when social networks are divided along partisan lines.

- ▶ Nodes: ?
- ▶ Link types: ?
- ▶ Link functions: ?

Possibly relevant are social ties through which political views are discussed, exchanged, reinforced.

- ▶ Nodes: ?
- ▶ Link types: ?
- ▶ Link functions: ?

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“To successfully defend their wealth[...], oligarchs promote to the position of legislators their own loyal clients, that is, trustworthy individuals who are personally related to their patrons, deeply invested in the relationships with them, and unlikely to cheat. These loyalists provide oligarchs with consistent representation in the legislature and work towards the adoption of legislation that is as favorable to their patrons’ economic interests as possible.” (Nitsova 2026)

- ▶ Nodes: ?
- ▶ Link types: ?
- ▶ Link functions: ?

Possibly relevant social ties are pre-existing personal connections based on trust and dependence, allowing the transmission of influence.

Designing the Study: Measurement

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Assume the relevant links are relationships that enable influence and recruitment.

1. Belong to the same online political group/forum
2. Regularly engage in message exchanges on the same online political group/forum

Assume the relevant links are social relationships through which political views are exchanged or reinforced.

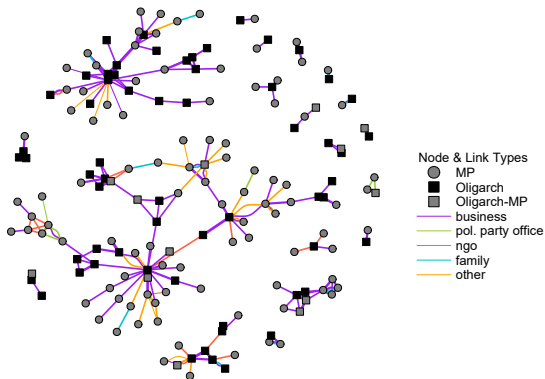
1. Report being friends
2. Report discussing politics with one another

At the level of theory:

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- ▶ oligarch: any individual in the top-100 richest people for 2012-2014
- ▶ MP: any individual who served as MP in the 8th Ukrainian Parliament (2014-2019) and 9th Ukrainian Parliament (2019-2022)
- ▶ oligarch-MP: any oligarch who also served as MP

- ▶ affiliated with the same business company...
- ▶ affiliated with the same NGO...
- ▶ affiliated with the same political party organization at the branch/office/leadership level...
... at the same time for any period prior to the respective election
- ▶ belong to the same family
- ▶ other (residual category for other personal connections)



Verkhovna Rada 8 Network: 172 nodes (61 oligarchs, 98 MPs, 13 oligarch-MPs) and 235 multiple edges. Density: 0.0137.

Why not measure and add the following links:

- ▶ political party membership or prior parliamentary faction membership instead of political party branch/office/leadership;
- ▶ university affiliation;
- ▶ friendship?

1. State your **network theory**:

2. Translate the theory into **measurement**:

- Discuss in groups of 3-4. Help each other identify anything that is still underspecified.

Designing the Study: Data Collection

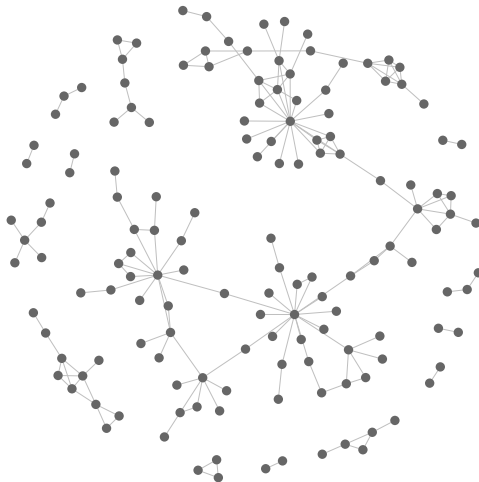
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The University of Manchester

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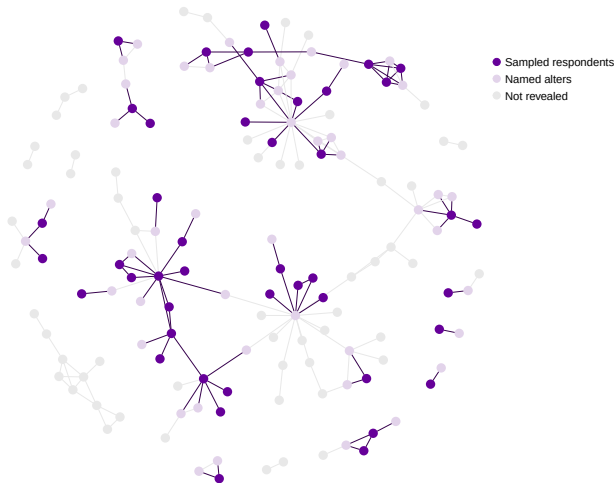
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- ▶ Sampling network data is different from ordinary survey sampling:
 - ▶ sampling respondents also determines which links and which parts of the network structure become observable.
- ▶ Three broad approaches
 - ▶ Census/bounded-network design: attempt to survey all respondents in a defined population.
 - ▶ Probability sampling: sample respondents from a known population.
 - ▶ Link-tracing sampling: start with seeds (small initial set of respondents) and reach additional respondents through reported links.

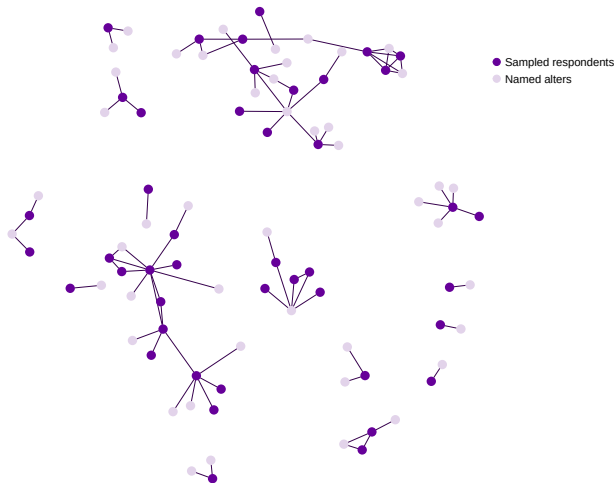
Ground truth, known population



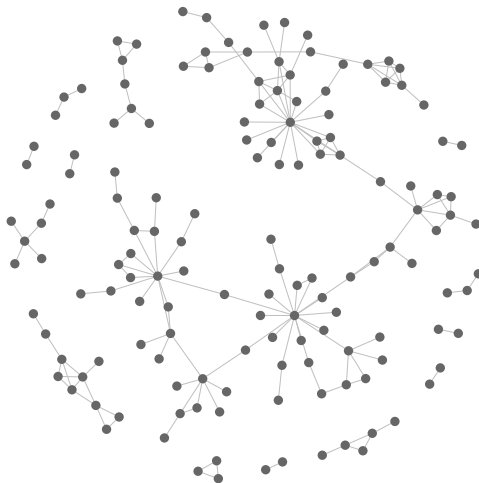
Simple random sampling



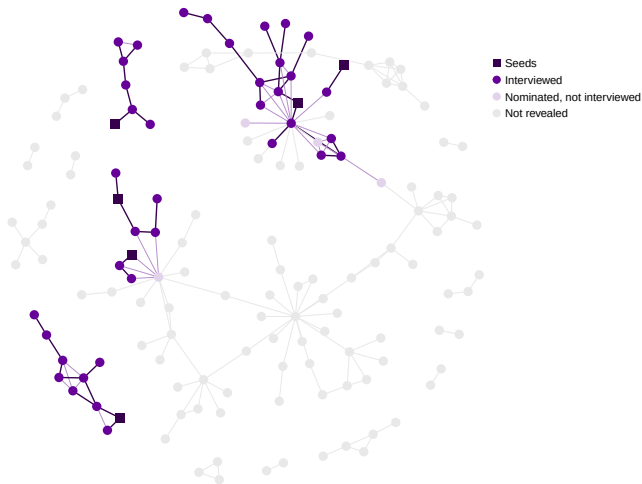
Simple random sampling



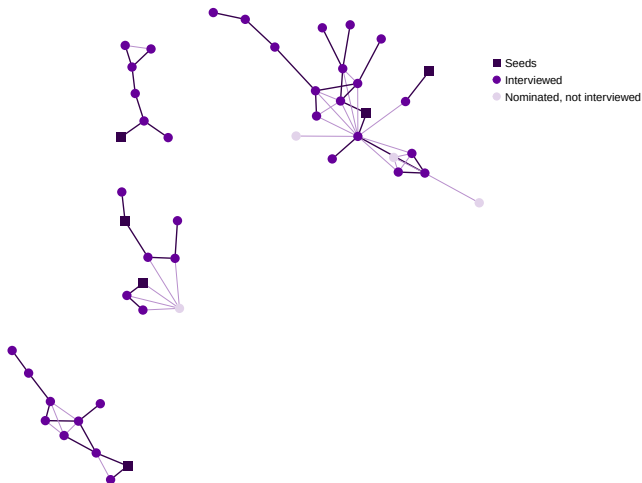
Ground truth, unknown population



Link-tracing sampling



Link-tracing sampling



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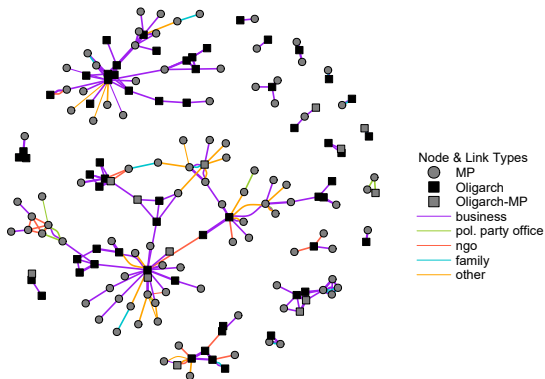
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Other Design Considerations

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How would you actually obtain the network data you need?

- Discuss in groups of 3-4.

